Offline Louisiana Insect Classifier

Bottom Line Up Front: An offline insect classifier for the state of Louisiana is to be created. The system will be able to run offline in an embedded system, allowing use in precision pest management solutions.

Background & Motivation

An insect identification system for the state of Louisiana would bring many benefits to the world of agriculture. This system will also be able to work offline, allowing systems without internet access to still participate in identification. Traditionally, such identification systems have been Online (limiting actual use in the field) and for the entire US (limiting accuracy).

Data Sources

iNaturalist is a gigantic database of pictures of identified insects, and consists of over 500k different species. In order to complete this project, I will be collecting and receiving a list of Louisiana-specific insects, cross examining the picture labels with the received list, and extracting all matches. There is likely to be a bias toward common insects, so dataset normalization is needed.

A Community for Naturalists · iNaturalist

Possible Technical Approach

Once the insect dataset has been curated, the problem becomes a simple classification issue. In the past, architectures such as ResNet have worked to get to accuracies up to 92 percent. As such, I hypothesize that a louisiana-specific model will be able to achieve a higher accuracy using the same architecture, or using a transfer-learning approach. Another approach is to use a Vision Transformer instead of a ResNet. However, the offline constraint will limit the size of the model. Thus, a model can be trained online, but will have to be compressed to run inference offline. This can maybe be achieved via LoRa or other entropy-based approaches.

Impact & References

If this project were to be successful, embedded devices such as Precision Pest Management robots would have the ability to identify insects and administer pesticides, potentially increasing the yield of a farmer's crop. This will potentially increase a farmer's profits, which will spread out through the economy.

I will be working with Dr. Grijalva of LSU Ag Engineering for a more thorough solution, but creation of this classifier is the first step. In addition, my company, FarmSmart, can then host online versions of this model and have the offline versions downloaded via our app, and we can extend this model to include all US states.